DTFT:

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \longrightarrow \text{DTFT}$$

$$x[n] = \frac{1}{2\pi} \int_{\langle 2\pi \rangle} X(\Omega) e^{j\Omega n} \, d\Omega \longrightarrow \text{Inverse DTFT}$$

Properties:

1) $x[n] \longrightarrow \boxed{\text{DTFT}} \longrightarrow X(\Omega)$ } Linear system & one-one

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$$

2) $X(\Omega)$ is periodic with period 2π.

Pf: $X(\Omega + 2\pi) = \sum_n x[n] e^{-j(\Omega+2\pi)n} = \sum_n x[n] e^{-j\Omega n} \cdot 1 = X(\Omega)$

3) $x[n]$ is real $\Rightarrow X(-\Omega) = X^*(\Omega)$

Pf: $\left.\begin{aligned} X(-\Omega) &= \sum_{n=-\infty}^{\infty} x[n] e^{j\Omega n} \\ X^*(\Omega) &= \sum_{n=-\infty}^{\infty} x^*[n] e^{j\Omega n} = \sum_{n=-\infty}^{\infty} x[n] e^{j\Omega n} \end{aligned}\right\} \Rightarrow X(-\Omega) = X^*(\Omega)$

$\therefore$ $|X(-\Omega)|$ is symmetric

$\angle X(\Omega)$ is antisymmetric

4) $x[-n] \longleftrightarrow X(-\Omega)$

Pf: $\sum_{n=-\infty}^{\infty} x[\underbrace{-n}_{n'}] e^{-j\Omega n} = \sum_{n'=-\infty}^{\infty} x[n'] e^{j\Omega n'} = X(-\Omega)$

5) $x[n]$ is real & even $\Rightarrow X(\Omega)$ is real & even

Pf: $\left.\begin{aligned} &x[n] \text{ is real} \Rightarrow X(-\Omega) = X^*(\Omega) \\ &x[-n] = x[n] \Rightarrow \underline{X(-\Omega) = X(\Omega)} \end{aligned}\right\} \Rightarrow X(\Omega) = X^*(\Omega)$ $\therefore X(\Omega)$ is real

$\left.\begin{aligned} &\Rightarrow X(\Omega) = X^*(\Omega) \\ &\therefore X(\Omega) \text{ is real} \end{aligned}\right\}$ $\therefore X(\Omega)$ is real & even

$$x[-n] = x[n] \Rightarrow \underbrace{X(-\Omega) = X(\Omega)}_{\text{Even}} \quad \left\{ \begin{array}{l} \Rightarrow X(-\Omega) = X(\Omega) \\ \therefore X(\Omega) \text{ is real} \end{array} \right\} \text{is real \& even}$$

6) $x[n-n_0] \longleftrightarrow X(\Omega)e^{-j\Omega n_0}$ } Time shifting

Pf: $$\sum_{n=-\infty}^{\infty} x[n-n_0]\, e^{-j\Omega n} = \sum_{n'=-\infty}^{\infty} x[n']\, e^{-j\Omega(n'+n_0)} \quad , n-n_0 = n'$$

$$= \sum_{n'=-\infty}^{\infty} x[n']\, e^{-j\Omega n'} \cdot e^{-j\Omega n_0}$$

$$= X(\Omega)\, e^{-j\Omega n_0}$$

7) $e^{j\Omega_0 n} x[n] \longleftrightarrow X(\Omega - \Omega_0)$ } Freq. shifting

Pf: $$\sum_{n=-\infty}^{\infty} x[n] e^{j\Omega_0 n} e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} x[n] e^{-j(\Omega-\Omega_0)n} = X(\Omega - \Omega_0)$$

8) $y[n] = x[n] * h[n] \longleftrightarrow Y(\Omega) = X(\Omega)H(\Omega)$

Pf: $$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] = \sum_{k=-\infty}^{\infty} x[k] \frac{1}{2\pi} \int_0^{2\pi} H(\Omega) e^{-j\Omega k} \cdot e^{j\Omega n}\, d\Omega$$

$$= \frac{1}{2\pi} \int_{\langle 2\pi \rangle} H(\Omega)\, e^{j\Omega n} \underbrace{\sum_{k=-\infty}^{\infty} x[k]\, e^{-j\Omega k}}\, d\Omega$$

$$Y[n] = \frac{1}{2\pi} \int_{\langle 2\pi \rangle} \underbrace{H(\Omega) X(\Omega)}_{Y(\Omega)} e^{j\Omega n}\, d\Omega$$

$\Rightarrow Y(\Omega) = X(\Omega)H(\Omega)$ $\therefore$ Proved.

8) $$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int |X(\Omega)|^2 d\Omega$$ } Parseval's Relat^n

Pf: $$\sum_{n=-\infty}^{\infty} x[n]\, x^*[n] = \sum_{n=-\infty}^{\infty} x[n] \frac{1}{2\pi} \int_{\langle 2\pi \rangle} X^*(\Omega) e^{-j\Omega n}\, d\Omega$$

$$= \frac{1}{2\pi} \int_{\langle 2\pi \rangle} X^*(\Omega) \underbrace{\sum_{n=-\infty}^{\infty} x[n]\, e^{-j\Omega n}}_{X(\Omega)}\, d\Omega$$

$$= \frac{1}{2\pi}\int_{\langle 2\pi\rangle} X(\Omega) \underbrace{\sum_{n=-\infty}}_{X(\Omega)}$$

$$= \frac{1}{2\pi}\int_{\langle 2\pi\rangle} |X(\Omega)|^2 \, d\Omega$$

9) $nx[n] \longleftrightarrow j\dfrac{dX(\Omega)}{d\Omega}$

Pf: $X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$

$$\frac{dX(\Omega)}{d\Omega} = \sum_{n=-\infty}^{\infty} x[n](-jn) e^{-j\Omega n}$$

$$= -j\sum_{n=-\infty}^{\infty} n x[n] e^{-j\Omega n}$$

$$\Rightarrow \underbrace{j\frac{dX(\Omega)}{d\Omega}}_{Y(\Omega)} = \sum_{n=-\infty}^{\infty} \underbrace{nx[n]}_{y[n]} e^{-j\Omega n} \iff nx[n] \longleftrightarrow j\frac{dX(\Omega)}{d\Omega}$$

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10) $y[n] = x_1[n]\, x_2[n] \longleftrightarrow \underbrace{Y(\Omega) = \frac{1}{2\pi}\int_{\langle 2\pi\rangle} X_1(\tau) X_2(\Omega-\tau)\, d\tau}_{\text{Periodic Convol}^n}$

Pf: $\frac{1}{2\pi}\int Y(\Omega) e^{j\Omega n} d\Omega = \frac{1}{2\pi}\int \frac{1}{2\pi}\int X_1(\tau) X_2(\Omega-\tau)\, d\tau\, e^{j\Omega n} d\Omega$

$$= \frac{1}{2\pi}\int X_1(\tau)\, d\tau \underbrace{\frac{1}{2\pi}\int X_2(\Omega-\tau) e^{j\Omega n} d\Omega}_{x_2[n] e^{jn\tau}}$$

$$= \frac{1}{2\pi}\int X_1(\tau)\, x_2[n]\, e^{jn\tau} d\tau$$

$$= x_2[n] \underbrace{\frac{1}{2\pi}\int X_1(\tau) e^{jn\tau} d\tau}_{x_1[n]}$$

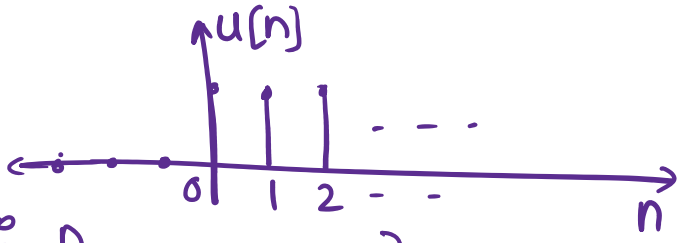
$y[n] = x_1[n]\, x_2[n]$ ∴ Proved.

$y[n] = x_1[n]\,x_2[n]$ · Proved.

$$\sum_{k=-\infty}^{n} x[k] \longleftrightarrow X(\Omega)\,U(\Omega) \qquad U(\Omega) = ?$$

$$\Downarrow$$

$$x[n] * u[n]$$



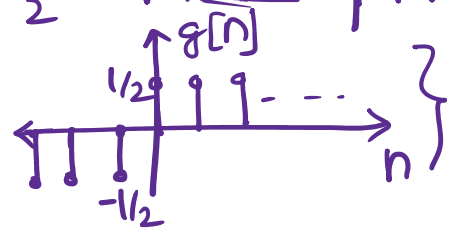
$$U(\Omega) = \sum_{n=0}^{\infty} e^{-j\Omega n} = \sum_{n=0}^{\infty} \alpha^n \quad , |\alpha| = 1 \quad \Big\} \therefore \text{Not converging.}$$

$$u[n] = f[n] + g[n] \quad ,$$

$$f[n] = \frac{1}{2}$$

$$f[n] = \frac{1}{2} \quad \forall n$$

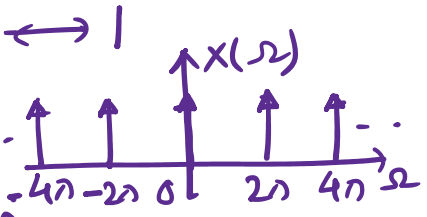
$$g[n] =$$



$$\Big\} \; g[n] - g[n-1] = \delta[n]$$

Let $x[n] = 1 \longleftrightarrow X(\Omega) = ?$ $\qquad \delta[n] \longleftrightarrow 1$

$$X(\Omega) = 2\pi \sum_{l=-\infty}^{\infty} \delta(\Omega - 2\pi l)$$



$$\frac{1}{2\pi}\int X(\Omega)\, e^{j\Omega n}\, d\Omega = \frac{1}{\cancel{2\pi}} \int_{-\pi}^{\pi} \cancel{2\pi} \sum_{l=-\infty}^{\infty} \delta(\Omega - 2\pi l)\, e^{j\Omega n}\, d\Omega$$

$$= \int_{-\pi}^{\pi} \delta(\Omega)\, e^{j\Omega n}\, d\Omega = 1 \quad ✓$$

$$\therefore \frac{1}{2} \longleftrightarrow \pi \sum_{l=-\infty}^{\infty} \delta(\Omega - 2\pi l) \quad \Big\} = F(\Omega)$$

$$g[n] - g[n-1] = \delta[n] \longleftrightarrow G(\Omega) - G(\Omega)\, e^{-j\Omega} = 1$$

$$\Rightarrow G(\Omega) = \frac{1}{1 - e^{-j\Omega}} \quad ✓$$

$$\Rightarrow U(\Omega) = F(\Omega) + G(\Omega)$$

$$= \frac{1}{1 - e^{-j\Omega}} + \pi \sum_{l=-\infty}^{\infty} \delta(\Omega - 2\pi l)$$

$$\therefore \sum^{n} x[k] \longleftrightarrow \frac{X(\Omega)}{\;} + \pi X(0) \sum_{l=-\infty}^{\infty} \delta(\Omega - 2\pi l)$$

$$\therefore \sum_{k=-\infty}^{n} x[k] \longleftrightarrow \frac{X(\Omega)}{1-e^{-j\Omega}} + \pi X(0) \sum_{l=-\infty}^{\infty} \delta(\Omega - 2\pi l)$$